MILITARY SPECIFICATION

WINCH, DRUM, POWER-OPERATED, ELECTRIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers electric drum type winches for installation on Naval ships.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

FF-S-85 - Screw, Cap, Slotted and Hexagon Head.
FF-S-86 - Screw, Cap, Socket-Head.
FF-S-92 - Screws, Machine: Slotted, Cross-Recessed or Hexagon Head.
HH-L-361 - Linings; Brake (Automotive Use).
TT-E-490 - Enamel, Silicone Alkyd Copolymer, Semigloss, Exterior.
TT-P-57 - Paint, Zinc Yellow-Iron Oxide-Base Ready-Mixed.

MILITARY

MIL-B-857 - Bolts, Nuts, and Studs.
MIL-C-2212 - Controllers, Electric Motor, A.C. or D.C., and Associated Switching Devices, Naval Shipboard.
MIL-P-3184 - Packaging of Machinery: Deck and Vehicle Mounted With Associated Equipment and Repair Parts.
MIL-B-16392 - Brakes, Magnet, Naval Shipboard.
MIL-M-17060 - Motors, 60 Hertz, Alternating Current, Integral Horsepower (Shipboard Use).
MIL-M-17413 - Motors, Direct-Current, Integral H.P., (Shipboard Use).
MIL-S-22473 - Sealing, Locking, and Retaining Compounds; Single-Component.
MIL-I-24137 - Iron Castings, Nodular Graphitic (Ductile Iron) and Nodular Graphitic (Corrosion Resisting, Austenitic, Low Magnetic Permeability) (For Shipboard Application).

STANDARDS

MILITARY

MS15004 - Fittings, Lubrication (Hydraulic) Surface Check, 1/4-28 Taper Threads, Nickel-Copper Alloy, Type IV.
MS15005 - Fittings, Lubrication Throat or Surface Check, 1/8 Pipe Threads, Nickel-Copper Alloy, Type V.

DRAWING

MILITARY

S2601-860303 - Capstan and Gypsy Heads.

PUBLICATION

MILITARY

NAVSHIPS 0901-190-0002, Chapter 9190 - Preservation of Ships In Service.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS
Handbook H28 - Screw Thread Standards For Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

STEEL STRUCTURES PAINTING COUNCIL (SSPC)
Painting Manual, Volume 2, Systems and Specifications, Surface Preparation Specification No. 10 - Near White Blast Cleaning.

(Application for copies should be addressed to the Steel Structures Painting Council, 4400 5th Avenue, Pittsburgh, Pa. 15213.)

AMERICAN GEAR MANUFACTURER'S ASSOCIATION (AGMA)
AGMA Publications.

(Application for copies should be addressed to the American Gear Manufacturer's Association, 1330 Massachusetts Avenue, N.W., Washington, D.C. 20005.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material. Materials shall be suitable for the purpose intended. Materials for mechanical items shall be in accordance with either commercial or Government specifications unless otherwise specified in the contract or order.

3.1.1 If commercial specifications are used the bill of material shall list both the commercial specification and the equivalent Government specification.

3.1.2 The use of cast iron shall be limited to material conforming to MIL-I-24137.

3.1.3 Radiographing of steel castings is not required.

3.2 Winch. The winch shall consist of the specified drums and gypsy heads (see 6.1.1), mounted horizontally on the same shaft with necessary reduction gearing, electric motor, electric brake, control equipment, and all operating accessories necessary for operation, exclusive of power leads. The winch shall be self contained on a single rigid bedplate except for the control equipment which shall be suitable for remote installation. The arrangement of all winch components on the bedplate with respect to each other shall be the same for all winches.

3.2.1 Design objectives. The basic design objectives are that the equipment will meet the needs of the Naval service and that the final product will reflect the utmost in simplicity, provide maximum safety for shipboard personnel with regard to both design and operation, have maximum reliability consistent with the state of the art, and be easy to install and maintain. The equipment shall be designed to withstand intermittent use over long periods of time under Naval service conditions, without benefit of overhaul. The design shall include all possible features which will result in reliable and stable operation with reduced requirements for adjustment and alinement, reduced requirement for maintenance, and simplified maintenance, thus reducing requirements for skilled personnel.

3.2.1.1 Design of the winch shall be such as to eliminate hazards of parts becoming inoperative due to frequent soaking by salt spray and extreme atmospheric conditions occurring with the winch in an exposed position on the weather deck of the ship. The winch shall be suitable for operation between ambient temperature of 120° Fahrenheit (F) to 0°F and shall have a design life of not less than the number of hours specified (see 6.1.1).

3.2.2 Drums. The drums shall have barrel diameters, widths and storage capacities as specified (see 6.1.1). Each drum shall have flange depth to stow the specified length and size wire rope (see 6.1.1) with a minimum of 2 inches of clear flange above the top layer. Each drum shall be guarded as necessary to prevent the wire rope from jumping the drum.

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3.2.2.1 Storage capacity and average layer. Storage capacity and average layer of the drum shall be determined by the following formulas:

$$L = (A + B) \times A \times C \times F$$
$$D = B + A + d/2$$
$$TL = B + 2A - d$$

Where,

L = Capacity of drum in feet of wire.
A = Depth of wire spooled on drum.
B = Barrel diameter.
C = Drum width.
TL = Pitch diameter of the top layer.
D = Pitch diameter of average layer.
d = Wire diameter.
F = Spooling factor = $\frac{.262}{d^2}$ (for controlled winding) or $\frac{.210}{d^2}$ (for uncontrolled winding).

3.2.2.2 Ratchet and pawl. Each drum shall be equipped with a ratchet and pawl designed to hold the drum under a line pull equal to the winch rated load with the wire rope on the top layer. The pawl shall be heavily counter-balanced or spring controlled so as to fall out of engagement when the load on the pawl is relieved. This lever shall also be fitted with a locking pin, ratchet, or similar means which will positively secure the pawl in the "out" position.

3.2.2.3 Brake. Each drum shall be equipped with an externally contracting flat band brake operated by a hand wheel (minimum diameter of the hand wheel shall be at least 12 inches). The drum brake shall be designed to hold a static load of 150 percent of the winch rated line pull, with the wire rope on the top layer. The effort to set the brake to its specified static capacity shall be equal to or less than a total of 60 pounds applied on the hand wheel. The brake mechanism shall be designed to withstand a force of 200 pounds on the handwheel without creating excessive loads on the drum shaft and drum shaft bearings (supporting calculations shall be included with the stress calculations). The brake lining shall be in accordance with MIL-L-361 and the unit pressure shall be equal to or less than that recommended by the manufacturer for the particular application. Positive means shall be provided for holding the brake band clear of the drum when the brake is released. The brake drum and brake band shall be designed to afford maximum protection to personnel and to preclude the entrance of foreign matter under the brake band. Means shall be provided to permit inspection, repairs and adjustment for wear.

3.2.2.4 Drum clutch. Each drum shall be equipped with a positive type clutch capable of being engaged in less than 90 degrees of drum rotation. The clutch yoke lever shall extend beyond the movable clutch section and shall be equipped with a positive means of locking the clutch in both the engaged and disengaged positions. The clutch shall be visible to the operator so he can determine by visual examination when the clutch sections are in alinement and when they are engaged. The rigidity of the clutch lever shall preclude the operator from locking the handle in the engaged position without the clutch actually being engaged.

3.2.2.5 Rope lead. The wire rope shall lead from the top of the drum back over the electric motor. The winch design shall be such that the wire rope may be led from the first layer on the barrel of the drum over the electric motor at an angle of at least 7-1/2 degrees below the horizontal.

3.2.2.6 Guards. Guards shall be provided for all rotating shafts with speeds in excess of 100 revolutions per minute (r/min).

3.2.3 Gypsy heads. The gypsy heads shall be proportioned in accordance with Drawing S2601-860303 for the size of manila rope specified (see 6.1.1). The wearing surfaces of the gypsy heads shall be hardened to a minimum Brinell of 325 to prevent scoring when used with wire rope. If gypsy heads are required, the design of a single gypsy winch shall be such as to permit ready removal of the gypsy head in the field without the use of special tools and reassembly on the opposite end of the drum shaft without the substitution of any parts. Special tools are defined as those tools not listed in the Federal Supply Catalog (copies of this catalog may be consulted in the office of the Defense Contract Administration Service (DCAS)). The arrangement shall provide clear leads both fore and aft at each gypsy head location.

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3.2.4 Gearing. Gear drives shall be used throughout. Gearing shall be in accordance with AGMA publications. Gearing shall be either single or double helical type and shall have machine cut teeth. Spur gears may be used for winches rated 15 horsepower or less. Gearing shall be totally enclosed in an oiltight case and shall be lubricated by an oil bath. Each gear case shall be fitted with a drain which will permit complete draining of the gear case.

3.2.4.1 Inspection holes for gearing. Housings containing gears shall be provided with adequately covered openings in sufficient number to allow visual inspection of the teeth on as many gears as practicable. Such openings shall expose to view at least two gear teeth at a time. The openings should be at the meshing points of the gearing, but may be located elsewhere, if such an arrangement is impracticable.

3.2.4.2 Dipstick. A dipstick shall be provided for indicating the oil level in each gear case. The dipstick should be marked to indicate the "FULL" and "ADD OIL" levels.

3.2.5 Bearings. Roller or ball bearings shall be used for gear and drum shafts. Accessories and accessory mechanisms, such as drum pawls, clutches, brakes, and operating mechanisms shall use needle or roller bearings. In all applications of antifriction bearings, the assembly drawing of the winch shall bear a certificate signed by a responsible official of the bearing manufacturing company showing that with full consideration to the intended service requirements, the bearing applications proposed are approved by his company as to all details of mounting, loading, speed and lubrication.

3.2.6 Bedplate. The bedplate shall be of strength to maintain alinement of components without aid from the deck foundation. The thickness of the bedplate flanges shall at least equal the diameter of the hold down bolts used. Foundation bolt holes shall be drilled 1/16 inch undersize for reaming at installation to fit body bound bolts. The design shall eliminate pockets in which water can stand or collect. For design of controls, the top of the foundation may be assumed to be 6 inches above the deck.

3.2.7 Electric motors. Electric motors shall be in accordance with MIL-M-17413 or MIL-M-17060, as specified (see 6.1.1).

3.2.8 Electric motor controllers. Electric motor controllers shall be in accordance with MIL-C-2212, as specified (see 6.1.1).

3.2.9 Electric brake. The electric brake shall be in accordance with MIL-B-16392, as specified (see 6.1.1). The electric brake shall be mounted on the motor shaft or extension thereof. The brake shall be designed to stop and hold 150 percent winch rated load when lowering at rated speed (wire rope or average layer) and to withstand a static pull of 200 percent of the rated load without creepage (wire rope on the average layer).

3.2.10 Lubrication. Pressure grease fittings shall be in accordance with MS15004 or MS15005 and shall be provided for all bearings. Sliding clutches, unless operating in an oil bath, shall be provided with at least two grease fittings, 180 degrees apart.

3.2.11 Welding. Welded fabrication properly made and stress relieved is generally acceptable in lieu of castings.

3.2.12 Fasteners. Protective means shall be provided for securing keys, feathers, splines, nuts, screws, and collars to prevent their working loose. Bolts, cap screws and studs shall be in accordance with MIL-B-857, FF-S-85, FF-S-86 or FF-S-92. Fasteners with a nominal diameter of 1/2 inch or less and exposed with the unit in operating position shall be of corrosion-resistant material. Set screws, if used, shall have positive locking devices (locking compound conforming to MIL-S-22473 is acceptable). Screw threads shall be unified, coarse or fine thread series, in accordance with Handbook H28.

3.2.13 Size and weight. The winch shall be as compact and light as is consistent with adequate strength and rigidity. If required, maximum size and weight shall not exceed that specified (see 6.1.1).

3.2.14 Performance. The winch shall be designed to perform as specified (see 6.1.1). The line pull and line speed specified shall be the rating of the winch both hoisting and lowering with the wire rope on the average layer of the drum.

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3.3 Painting.

3.3.1 Surfaces to be painted shall be cleaned free of grease, oil, dirt, and other contaminants, prior to painting. Solvents or detergent washes shall be used to remove contaminants. Surfaces shall be wiped dry, or flushed clean with fresh water if appropriate, and dried before painting.

3.3.2 Before assembly, ferrous surfaces not intended to be bright shall be cleaned by abrasive blasting in accordance with SSPC SP-10 and be coated as follows:

(a) One coat of inorganic zinc silicate primer selected from NAVSHIPS 0901-190-0002, Chapter 9190.
(b) Apply one coat at 1.5 mils (minimum dry film thickness) of zinc yellow-iron oxide in accordance with TT-P-57, type II.

After assembly, primed surfaces shall be top coated with two coats of enamel in accordance with TT-E-490 (color-Navy haze gray). Total top coat dry film thickness shall be at least 5 mils.

3.3.3 Before assembly, nonferrous surfaces shall be primed with formula 117 (approximately 0.5 mils). After assembly primed surfaces shall be top coated with two coats of enamel in accordance with TT-E-490 (color-Navy haze gray). Total top coat film thickness shall be at least 5 mils.

3.4 Identification plates. Identification plates shall be made from either brass, nickel-copper alloy, or corrosion-resisting steel and shall be a minimum thickness of 0.03 inch. Stamping of letters, numerals, and other characters shall be at least 0.003 inch deep. Identification plates shall include type of winch, National stock number, rated line pull and line speed, manufacturer's identification, contract or order number, serial number, date of manufacture, and weight of winch.

3.5 Onboard repair parts. One set of repair parts shall be provided for one to five winches, of the same type and rating, installed on each ship. Two sets shall be provided for six to twenty units and three sets for twenty-one or more winches installed on each ship.

3.5.1 Mechanical. Each set of mechanical repair parts shall consist of the following items as applicable, in sufficient quantity for complete replacement for one winch:

(a) One set of bushing type bearings.
(b) One set of antifriction bearings.
(c) One set of seals and special packing.
(d) One set of brake linings and rivets.
(e) One complete set of springs.

3.5.2 Electrical. Electrical repair parts shall be furnished in accordance with MIL-C-2212, MIL-B-16392, MIL-M-17060, and MIL-M-17413 as applicable to the electrical items.

3.6 Technical data. The supplier shall prepare engineering drawings, technical manuals, and calculations in accordance with the data ordering documents included in the contract or order (see 6.1.2).

3.6.1 Drawings. In addition to the drawing content required by the data ordering document, the following unique technical features shall be included:

(a) Preliminary drawings shall consist of the following:
  (1) General arrangement drawing.
  (2) Assembly and subassembly drawings.
  (3) Electric motor, controller and brake drawings.
(b) The general arrangement drawing shall contain the following in addition to the assembly views:
  (1) A list of the mechanical repair parts furnished for the winch.
  (2) An index of all related drawings.
  (3) A notation of duty ratings for the winch.
  (4) Bedplate and foundation dimensions.
  (5) Calculated and actual weight and center of gravity

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3.6.2 Technical manuals. In addition to the general requirements covered by the data ordering document, the following unique technical features shall be included:

(a) Two preliminary copies shall be submitted to the command or agency concerned for review and for assignment of identification number. The preliminary manual shall contain a print of all of the drawings listed below:
(1) General arrangement drawing.
(2) Assembly and subassembly drawings in sufficient detail to adequately identify parts.
(3) Electric motor, electric brake and motor controller drawings.

3.6.3 Stress calculations. A complete set of stress calculations in duplicate of gears, bearings, shafts, keys, holding down bolts, and brake shall be included with the first submission of drawings for review.

3.6.3.1 Under rated conditions, the combined stress in any part shall not exceed 35 percent of the tensile yield point of the ductile materials used.

3.6.3.1.1 At no time shall the combined stress in any part of the winch exceed 70 percent of the tensile yield point when the gypsy is subjected to a horizontal pull equal to the breaking strength of the manila rope specified for use with this winch, when acting at midlength of the gypsy, through and normal to the axis of the shaft.

3.6.3.1.2 The combined stress in any part of the winch shall not exceed 70 percent of the tensile yield point when the drum is subjected to a pull equal to the breaking strength of the rope with the rope on the average layer with the drum pawl engaged or a shock load equal to 300 percent of the rated load with the rope on the top layer.

3.6.3.2 In stress calculations, the strength relationship shown in table I shall exist.

Table I - Strength relationship.

Design strength	Percent of tensile yield point
Direct shear	60
Torsional shear	65
Compression (bearing)	160

3.6.3.2.1 Other values may be used if substantiated by tests and accepted by the command or agency concerned.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection.

4.2.1 Examination. Each winch shall be examined to determine compliance with the requirements of this specification. Examination shall be conducted in accordance with table II. Only conforming winches shall be offered for delivery.

4.2.2 Tests. Each winch shall be tested for compliance with this specification. Only conforming winches shall be offered for delivery. Quality conformance tests shall consist of the tests specified in 4.3.1 through 4.3.5

4.3 Test methods.

4.3.1 No load. Each winch shall be operated under no load for a period of 1/2 hour in each direction of rotation. No abnormal heating (150°F) of any part shall occur.

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Table II - Examination.

Examination	Requirement paragraph
Guards provided for all rotating shafts with speeds in excess of 100 r/min.	3.2.2.6
Pawl is heavily counterbalanced or spring controlled to fallout of engagement when load on the pawl is relieved.	3.2.2.2
Roller or ball bearings used for gear shafts.	3.2.5
Roller or ball bearings used for drum shafts.	3.2.5
Needle or ball bearings used for accessories and accessory mechanisms such as drum pawls, clutches, brakes and operating mechanisms.	3.2.5
Pressure grease fittings provided for all bearings.	3.2.10
Pressure grease fittings provided for sliding clutch (not required if clutch operates in an oil bath).	3.2.10
Pressure grease fittings are in accordance with MS15004 or MS15005.	3.2.10
Each drum is equipped with a drum brake.	3.2.2.3
Each drum is equipped with a pawl and ratchet.	3.2.2.2
The diameter of the drum brake hand wheel is at least 12 inches.	3.2.2.3
The clutch yoke lever can be locked in both the engaged and disengaged positions.	3.2.2.4
The wire rope leads from the top of the drum back over the electric motor.	3.2.2.5
Inspection holes for inspecting gears provided.	3.2.4.1
Each gear case fitted with drain to permit complete draining of gear case.	3.2.4
Dipstick provided for measuring oil level in gear case.	3.2.4.2
Fasteners provided for securing keys, feathers, splines, nuts, screws and collars.	3.2.12
Identification plate provided.	3.4

4.3.2 Rated load. Each winch shall be operated, hoisting and lowering rated load at rated speed through a distance of at least 20 feet in each direction for five complete cycles. The load shall be stopped and held suspended for a period of 1 minute at the end of each cycle. The load shall be held suspended at all times.

4.3.3 Overload. Each winch shall demonstrate its ability to hoist and lower a load equal to 150 percent of rated load at no specified speed through a distance of at least 20 feet in each direction. The load shall be hoisted from a suspended position and lowered and stopped in a suspended position.

4.3.4 Drum brake. Each drum brake on each winch shall be subjected to a static pull equal to 150 percent of rated load. No slippage of the drum brake shall occur. The effort to set the drum brake shall be 60 pounds or less on the handwheel.

4.3.5 Electric brake. The electric brake of each winch shall be subjected to a static pull equal to 200 percent of the rated load for the winch (wire rope on the average layer) without slippage. Each electric brake shall demonstrate its ability to stop and hold a load equal to 150 percent of the rated load for the winch when lowered at rated speed under simulated power failure conditions. Stopping distance under this test shall be equal to or less than the distance the winch would payout when running at rated speed for 3 seconds. If the winch has more than one speed point the test should be conducted with the control in the highest speed point.

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4.4 Inspection of preparation for delivery. The packaging, packing and marking shall be inspected for compliance with section 5 of this document.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking. The equipment, accessories, manuals, and repair parts shall be preserved and packaged level A or C; packed level A, B or C and marked level A, B, or C as specified (see 6.1.1) in accordance with MIL-P-3184.

6. NOTES

6.1 Ordering data.

6.1.1 Procurement requirements. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) The number of drums and gypsy heads required (see 3.2).
(c) Design life of the winch (see 3.2.1.1).
(d) Drum requirements (see 3.2.2):
    (1) Barrel diameter.
    (2) Drum width.
    (3) Storage capacity.
    (4) Breaking strength and size of wire rope.
(e) Gypsy head requirements (see 3.2.3):
    (1) Manila rope size.
    (2) Breaking strength of rope.
(f) Electric motor requirements (see 3.2.7):
    (1) Specification.
    (2) Service.
    (3) Ambient temperature.
    (4) Voltage and phase rating.
    (5) Enclosure.
    (6) Duty.
    (7) Speed.
    (8) Type.
    (9) Design.
    (10) Winding.
    (11) Horsepower.
    (12) Overload (motors may be overloaded 25 percent for rated conditions).
(g) Electric motor controller requirements (see 3.2.9).
    (1) Specification.
    (2) Voltage and phase rating.
    (3) Enclosure.
    (4) Duty.
    (5) Ambient temperature.
    (6) Operation.
    (7) Type.
    (8) Performance.
    (9) Function.
    (10) Protection features.
    (11) Master switch.
(h) Electric brake requirements (see 3.2.9):
    (1) Specification.
    (2) Voltage.
    (3) Duty.
    (4) Ambient temperatures.
    (5) Type of enclosure.
    (6) Mounting.
(i) Maximum size and weight of winch, if a requirement (see 3.2.13).
    (1) Maximum width.
    (2) Maximum length (perpendicular to drum shaft).
    (3) Maximum height.
    (4) Maximum weight.

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(j) Performance data of the winch with wire rope on average layer of drum (see 3.2.14):
(1) Rated line pull.
(2) Rated line speed.
(3) No load line speed.
(k) Selection of applicable level of preservation, packaging, packing and marking required (see 5.1).

6.1.2 Contract data requirements. When this specification is used in a procurement invoking the data requirement clause of the Armed Services Procurement Regulations (ASPR) paragraph 7-104.9(n) and which incorporates a DD Form 1423 Contract Data Requirements List (CDRL), the data requirements identified below will be developed as specified in the cited Data Item Description (DID) and delivered in accordance with such CDRL. When the ASPR provisions are not invoked, the data specified below shall be delivered in accordance with the contract requirements.

	Specification paragraph	Data requirement	Service	Applicable DID	Options
(a)	3.6.1	Drawings	SH	UDI-E-23174	Categories E, G, and H, form 2, type II
(b)	3.6.2	Technical manuals	SH	UDI-M-23455	Paragraph 1(b)1
(c)	3.6.3	Calculations	SH	UDI-E-23213	---

(Copies of DID's required by the supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

6.2 THE MARGINS OF THIS SPECIFICATION ARE MARKED "#" TO INDICATE WHERE CHANGES (ADDITIONS, MODIFICATIONS, CORRECTIONS, DELETIONS) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS WAS DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

Preparing activity:
Navy - SH
(Project 3950-N172)

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OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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PRINCE GEORGES CENTER
HYATTSVILLE, MARYLAND 20782

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